Western Ghats Field Validator

Mobile Application for Ground-Truth Validation of Satellite-Derived Land Cover Classifications

> **Key Capabilities:** Offline-first architecture | CoRE Stack API integration (8 endpoints) | Forest vs plantation classification | GPS-based observation capture | JSON/CSV data export

1. Overview

The Western Ghats Field Validator is a mobile-first progressive web application that enables field researchers to validate satellite-derived LULC classifications through systematic ground-truth observations. It integrates CoRE Stack watershed APIs, Google Dynamic World, and custom forest typology datasets.

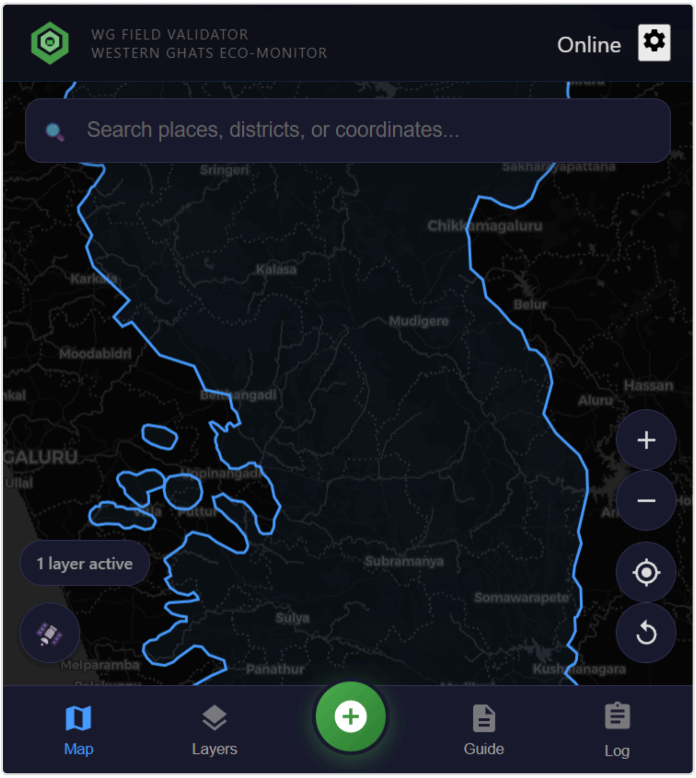

Figure 1: Main interface with Western Ghats boundary

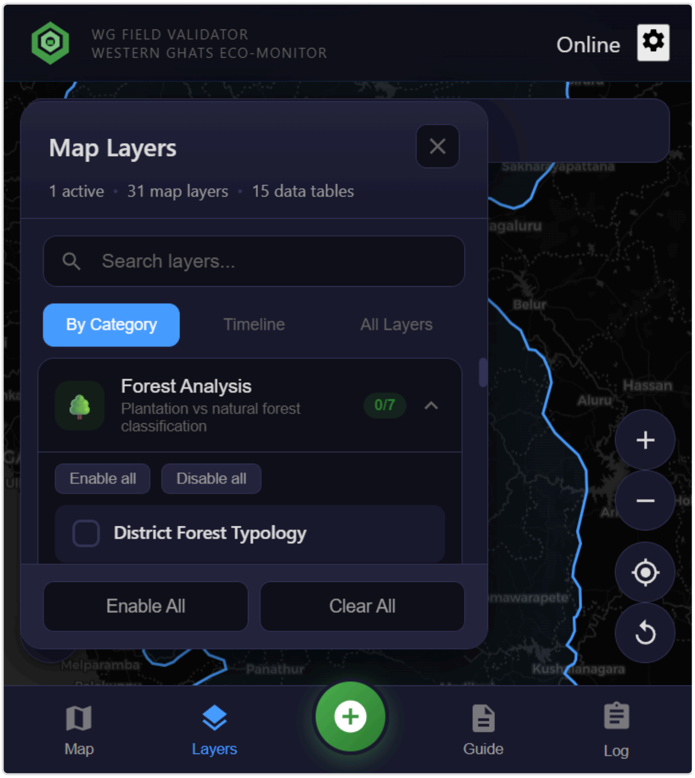

Figure 2: Layer panel with 31 map layers

Interface Features: Interactive map with street/satellite base layers | GPS location tracking | Layer toggle with active count | Bottom navigation (Map, Layers, Capture, Guide, Log)

2. Layer Management

The application organizes **31 map layers** and **15 data tables** across thematic categories:

Category	Description	Layers
Forest Analysis	Plantation vs natural forest classification	7
Land Cover Maps	Historical LULC from GLC-FCS30D (1985-2022)	16
Urban Expansion	Built-up area change detection	14
Dynamic World	Google Earth Engine LULC classification	2

Boundaries	Administrative boundaries (District, WG)	2
Watershed Data	Water balance, cropping intensity (CoRE Stack)	4

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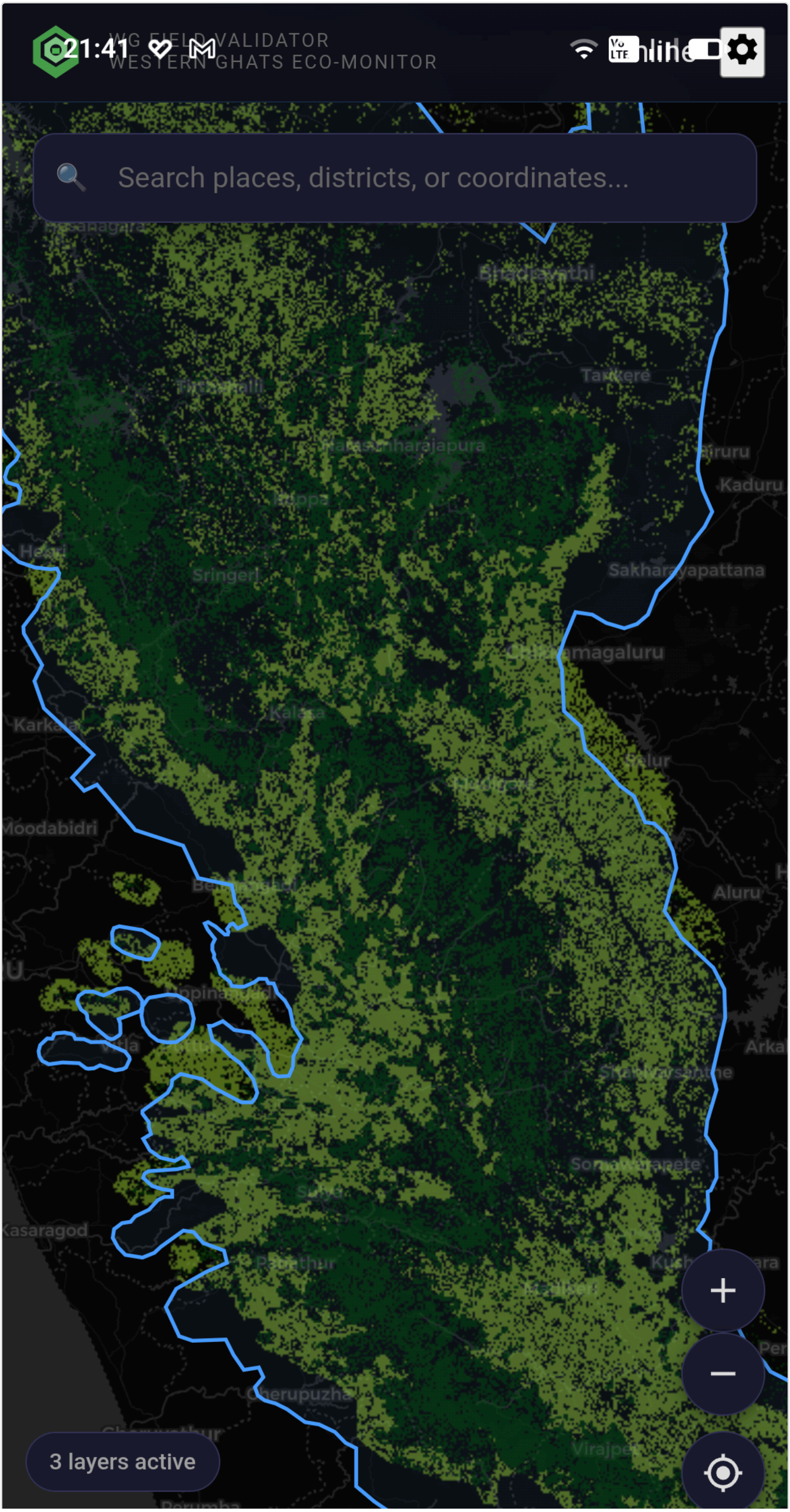
3. Forest vs Plantation Classification


21:41
WG FIELD VALIDATOR
WESTERN GHATS ECO-MONITOR
Search places, districts, or coordinates...
Bhadravathi
Tarikere
Tirthahalli
Narasimharajapura
Koppa
Kaduru
Sringeri
Sakharayapattana
Chikkamagaluru
Karkala
Kalasa
Belur
Moodabidri
Aluru
Arkal
Kasaragod
Cherupuzha
Virajpet
3 layers active

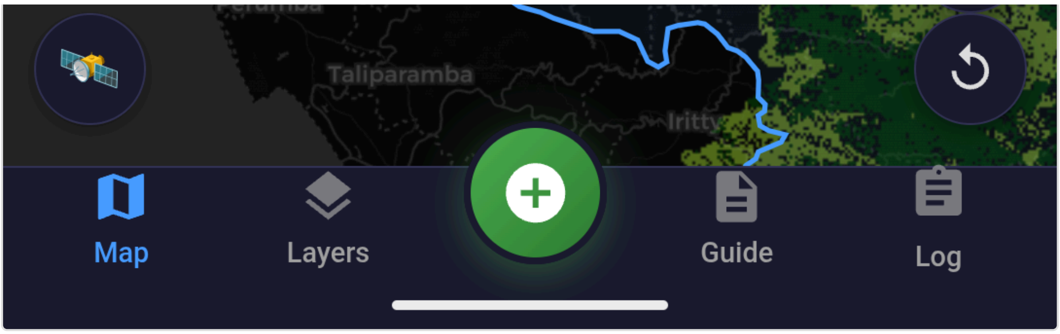


Figure 3: Forest typology showing natural forest (green) vs plantations (purple/magenta)

Classification Methodology

The forest typology distinguishes natural forests from plantations using multi-criteria analysis:

Natural Forest Indicators

- **Tree cover persistence:** Consistent cover 2000-2020 (Hansen GFC)
- **Canopy structure:** Multi-layered, heterogeneous heights
- **Spatial pattern:** Irregular boundaries, internal heterogeneity
- **Historical use:** No recent clearing/replanting cycles

Plantation Indicators

- **Spectral signatures:** Monoculture reflectance (rubber, teak, eucalyptus)
- **Row patterns:** Regular geometric planting arrangements
- **Age uniformity:** Even-aged stands from synchronized planting
- **Crop cycles:** Periodic harvesting in time-series

This distinction is critical for biodiversity assessments: natural forests support significantly higher species diversity than commercial plantations, despite both appearing as "forest" in conventional satellite classifications.

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4. Dynamic World and CoRE Stack Integration

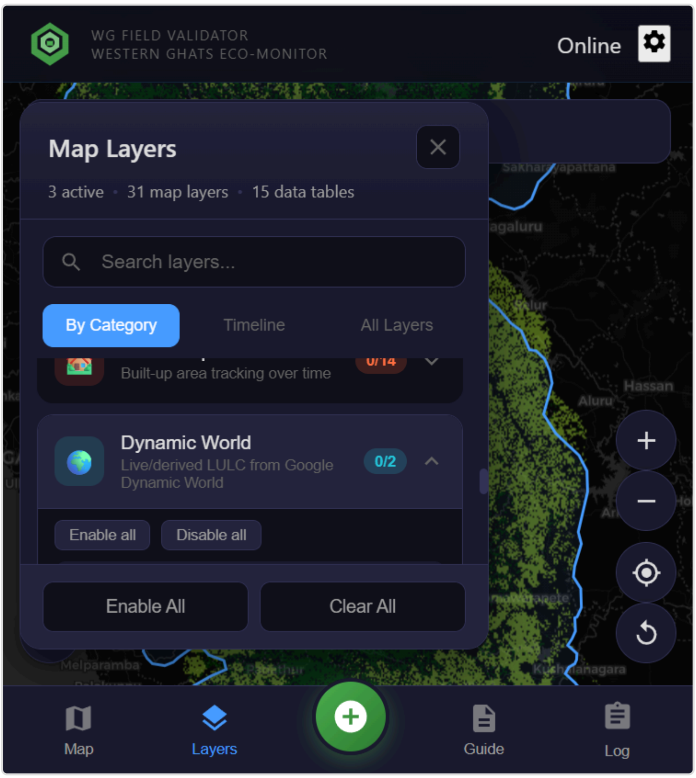

Figure 4: Dynamic World LULC options

Dynamic World LULC

Google's near real-time land cover with 9 classes: Water, Trees, Grass, Flooded Vegetation, Crops, Shrub/Scrub, Built-up, Bare Ground, Snow/Ice.

Options:

- Live GEE: Real-time via Earth Engine API
- Regional (2018-2025): Pre-processed composite

CoRE Stack API Endpoints

Endpoint	Data Provided
`get_admin_details_by_latlon`	State, District, Tehsil lookup from coordinates
`get_mwsid_by_latlon`	Micro-Watershed ID for location
`get_generated_layer_urls`	Available GIS layers for tehsil
`get_tehsil_data`	Comprehensive tehsil-level statistics
`get_mws_data`	Time series: Evapotranspiration, Runoff, Precipitation
`get_mws_kyl_indicators`	Know Your Landscape watershed indicators
`get_waterbodies_data_by_admin`	Waterbody inventory for admin unit
`get_mws_report`	MWS assessment report URL

All endpoints tested and documented in `docs/notebooks/corestack_api_tested.ipynb`.

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5. Field Observation Workflow

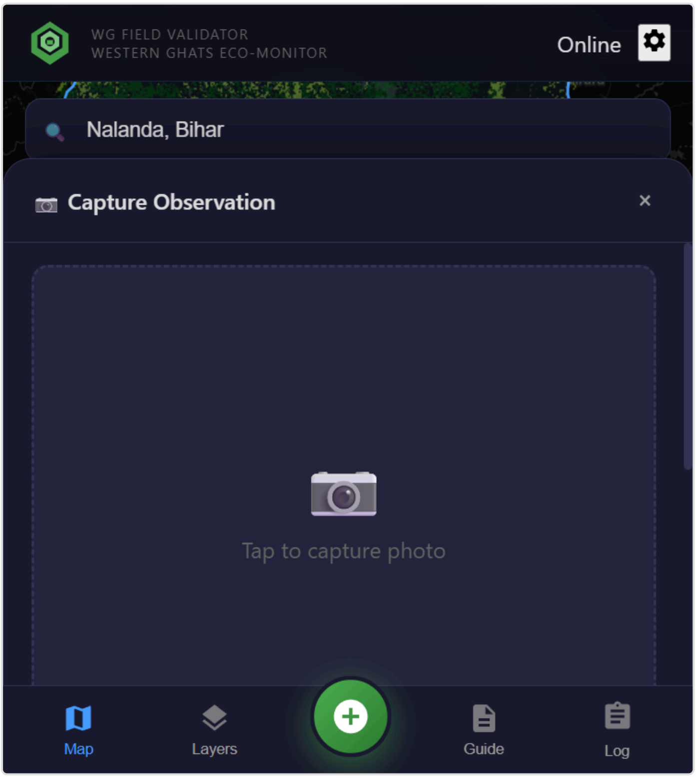

Figure 5: Observation capture interface

Capture Process

1. **Photo capture:** Camera or gallery selection
2. **GPS coordinates:** Automatic with accuracy indicator
3. **Layer values:** Active layer data at location
4. **Validation status:**
   - Match: Classification agrees with observation
   - Mismatch: Classification differs from ground truth
   - Unclear: Cannot determine accuracy
5. **Field notes:** Optional documentation

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6. Field Log and Data Export

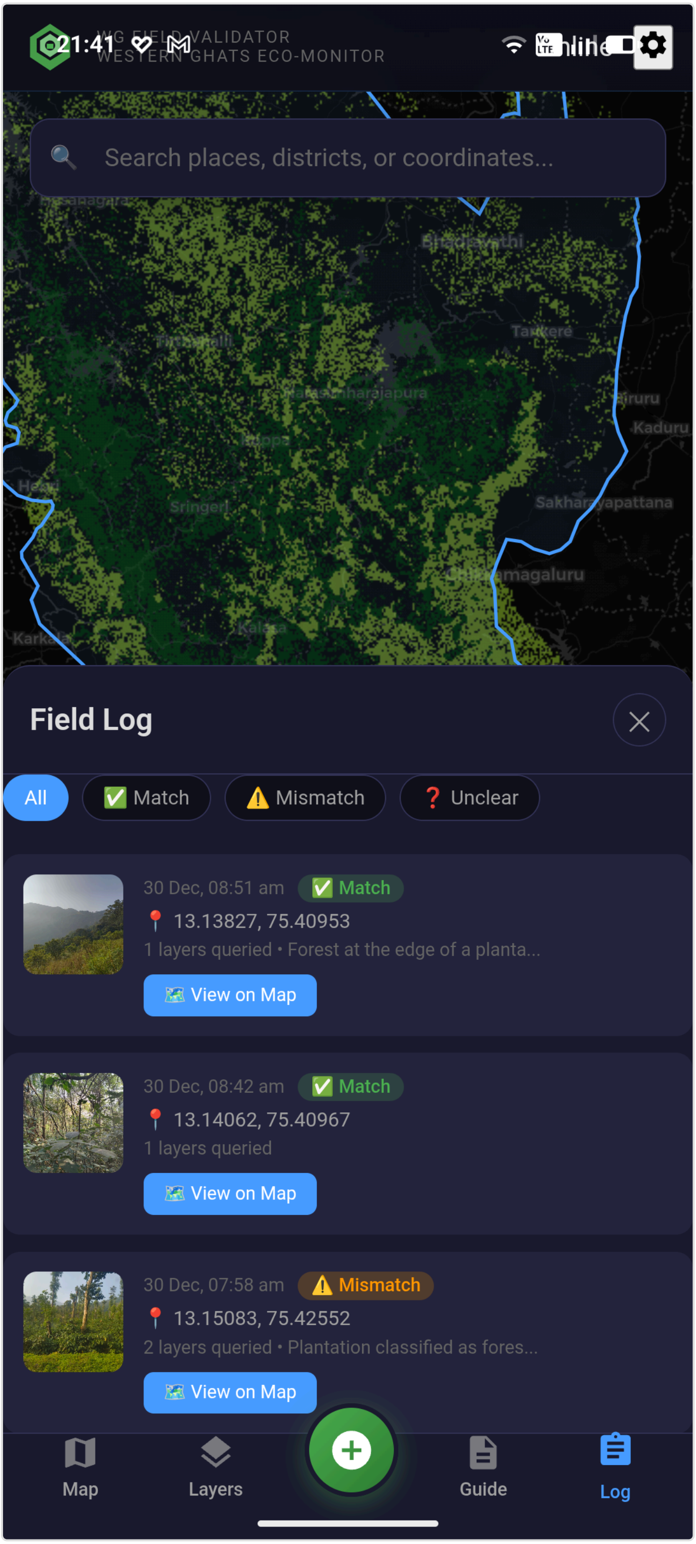

Figure 6: Field log with validation filtering

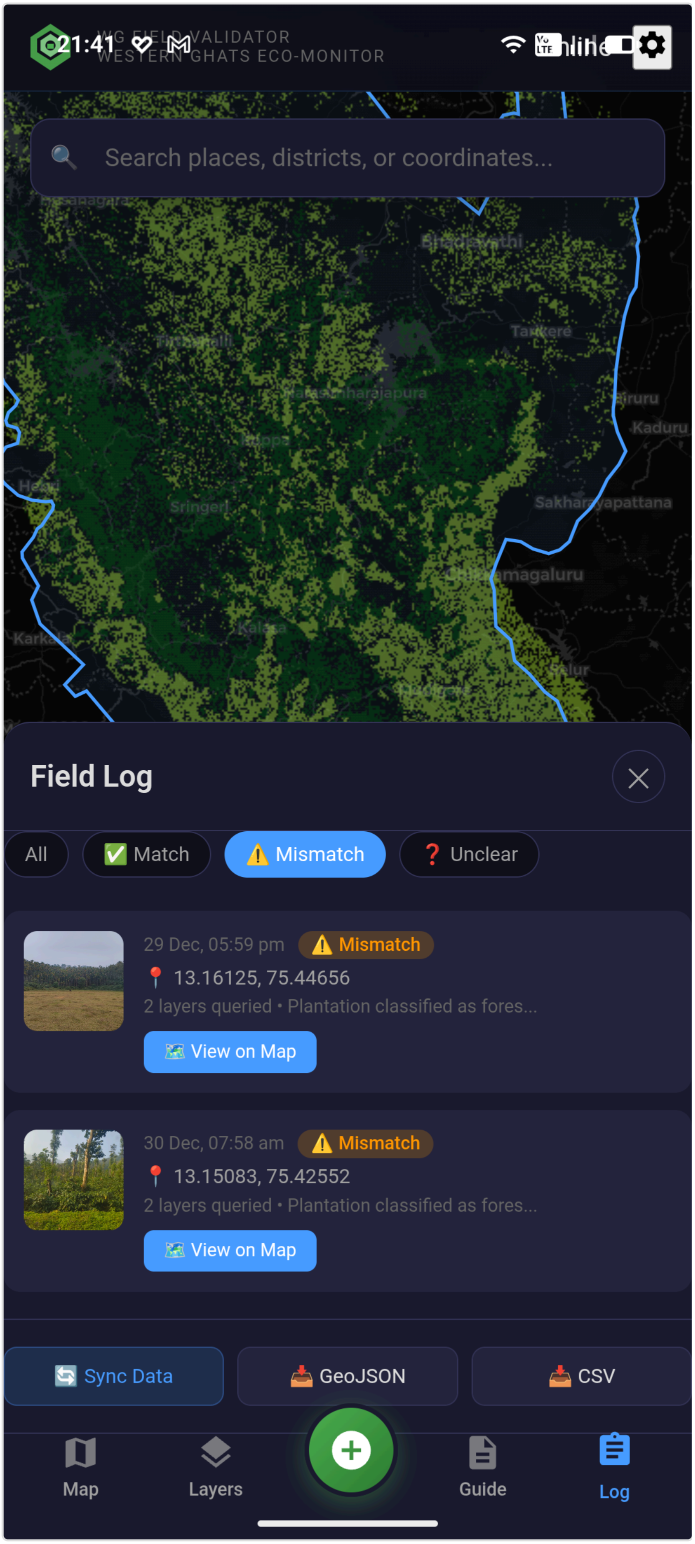

Figure 7: Data export options

Field Log Features:

- Chronological observation list with photo thumbnails
- Filter by validation status (All, Match, Mismatch, Unclear)
- View observation details and location on map

Export Capabilities:

- **JSON:** Complete data with metadata for programmatic analysis
- **CSV:** Tabular format for spreadsheets and GIS software
- **Offline storage:** Local IndexedDB with sync on connectivity

7. Additional Features

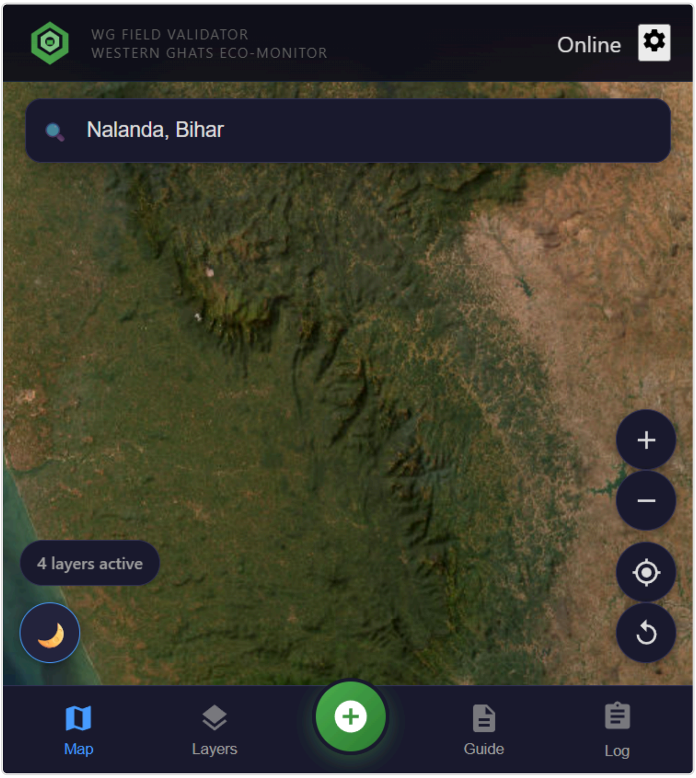

Figure 8: Satellite base map for verification

Base Map Options

- OpenStreetMap: Navigation context
- ESRI World Imagery: Visual verification
- Dark mode: Low-light conditions

Built-in Field Guide

- Getting started / GPS setup
- Navigation and map controls
- Layer management
- Observation capture workflow
- Offline mode operation

8. Technical Architecture and Benefits

Component	Technology
Frontend	React 18 + TypeScript + Vite
Mapping	MapLibre GL JS
Mobile	Capacitor (Android APK)
Storage	IndexedDB (offline-first)
APIs	CoRE Stack, OSM Nominatim

For Field Researchers

- Works offline in remote areas
- Automatic GPS with accuracy
- Multiple layer comparison

For Conservation Organizations

- Systematic ground-truth validation
- Plantation vs forest distinction
- Standardized methodology

Resources

Resource	Location
Sample field observations	`field-data/recovered-observations/`

CoRE Stack API notebook	`docs/notebooks/corestack_api_tested.ipynb`
Source code	MIT Licensed on GitHub